

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
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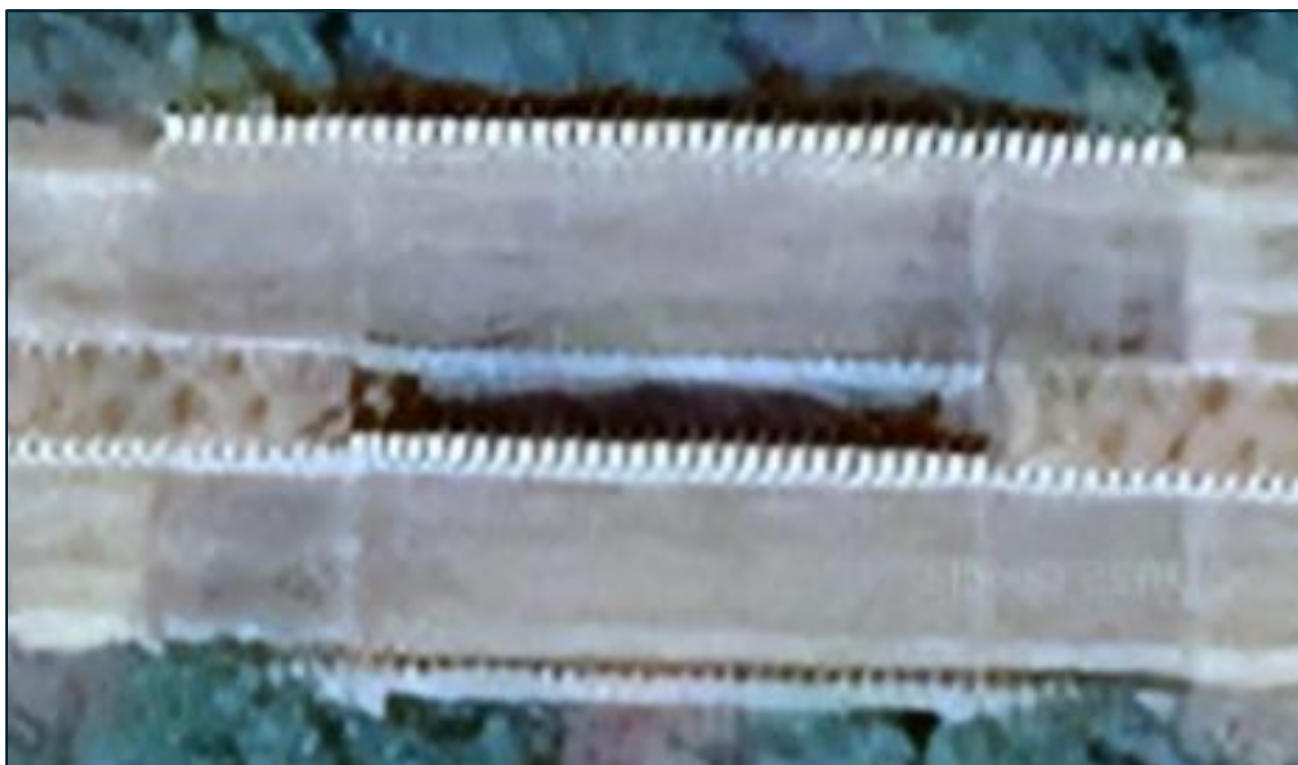
3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 929+030

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

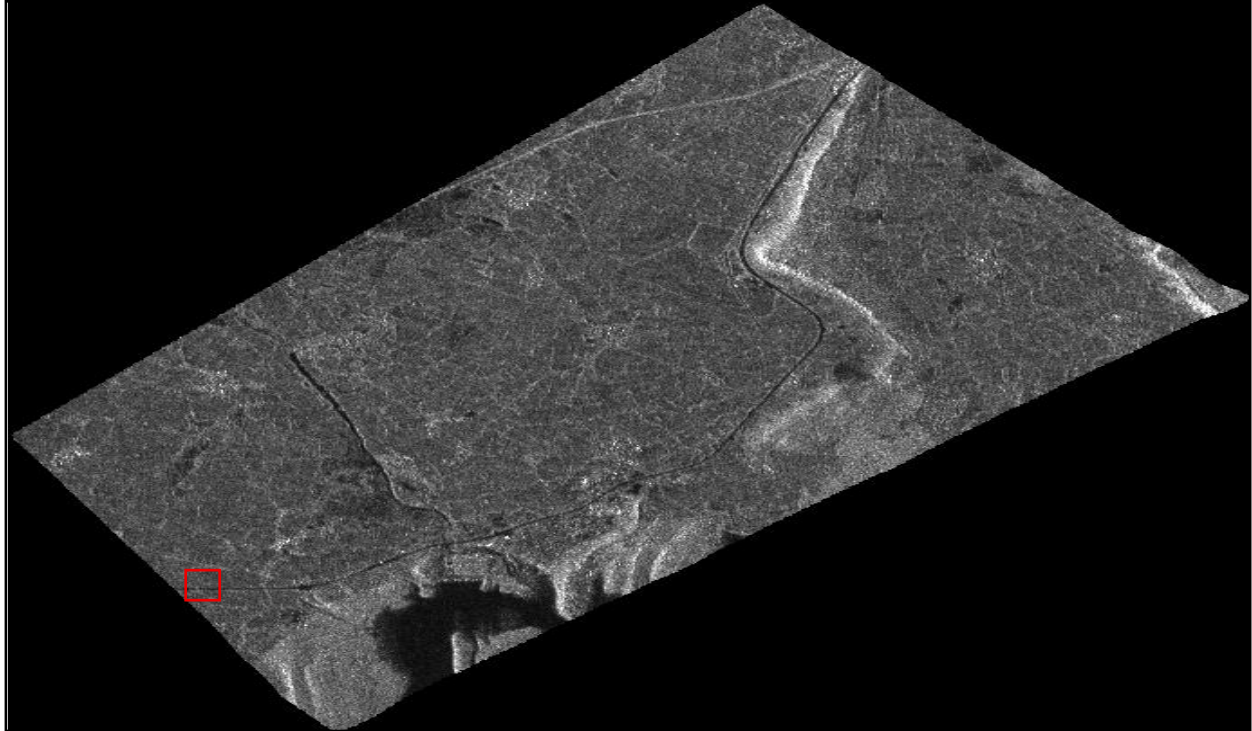
Flight Direction - DESCENDING

Polarization – VV+VH

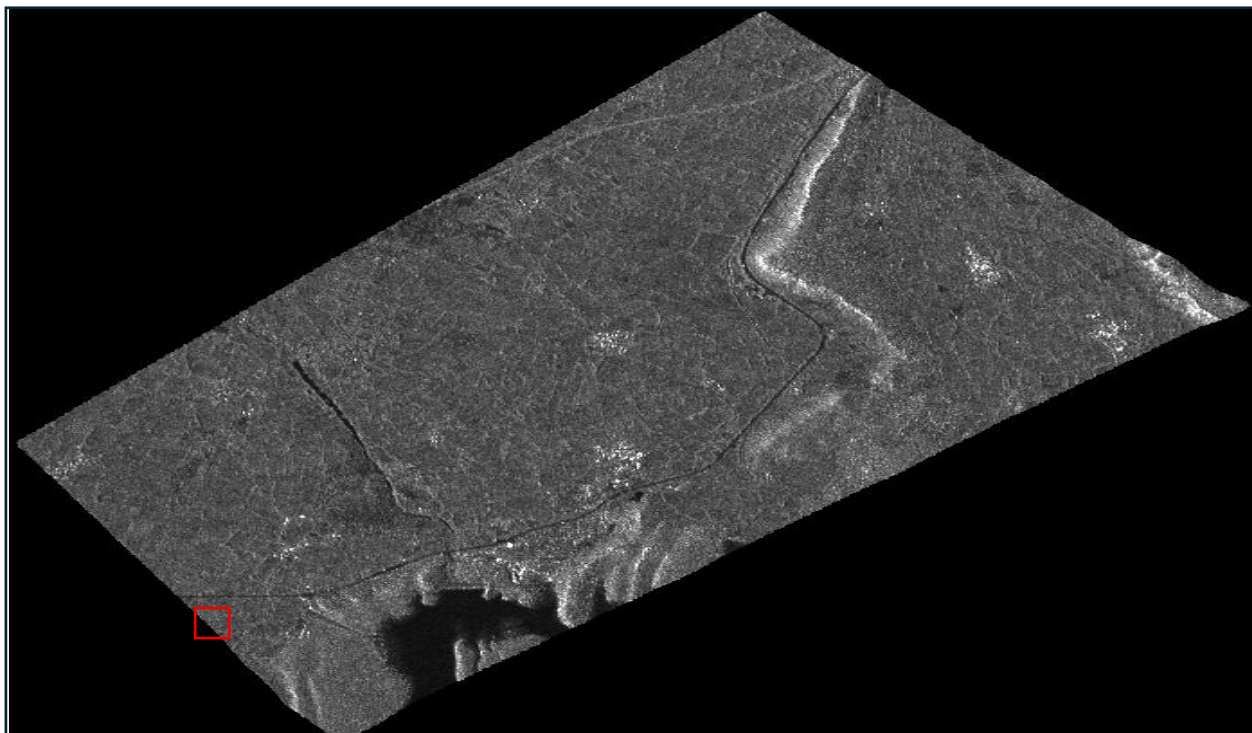
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Ortho rectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

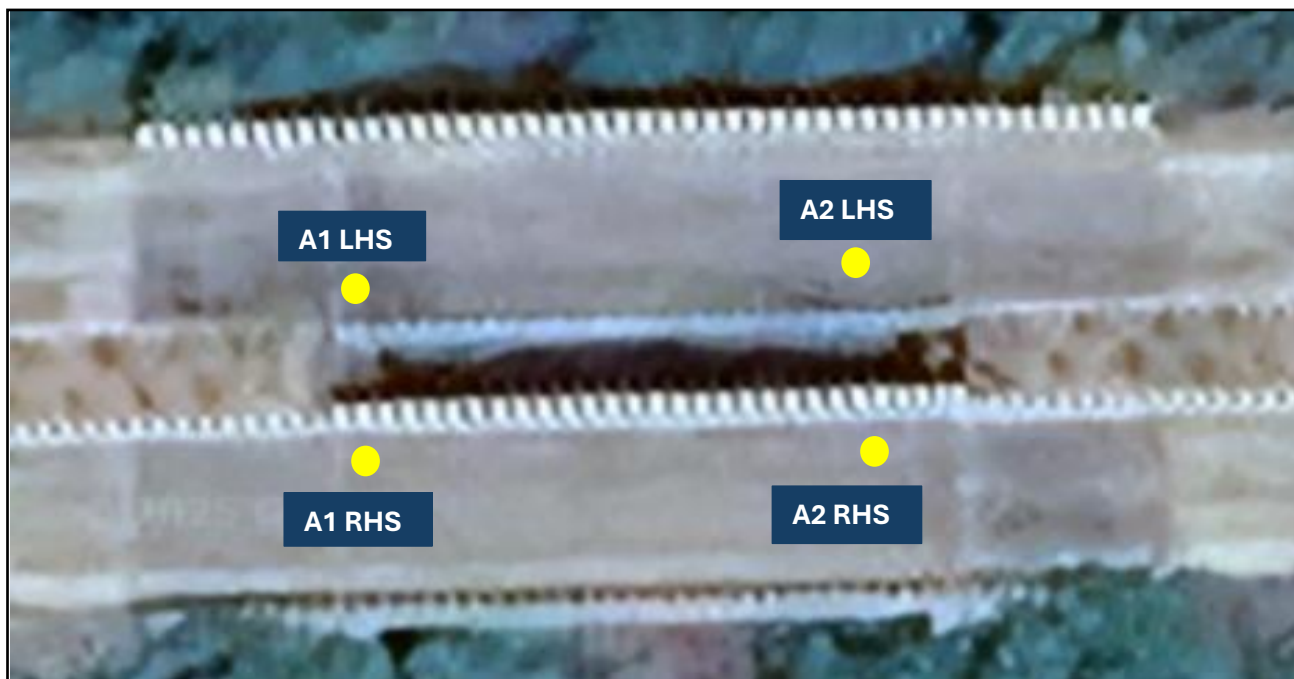


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

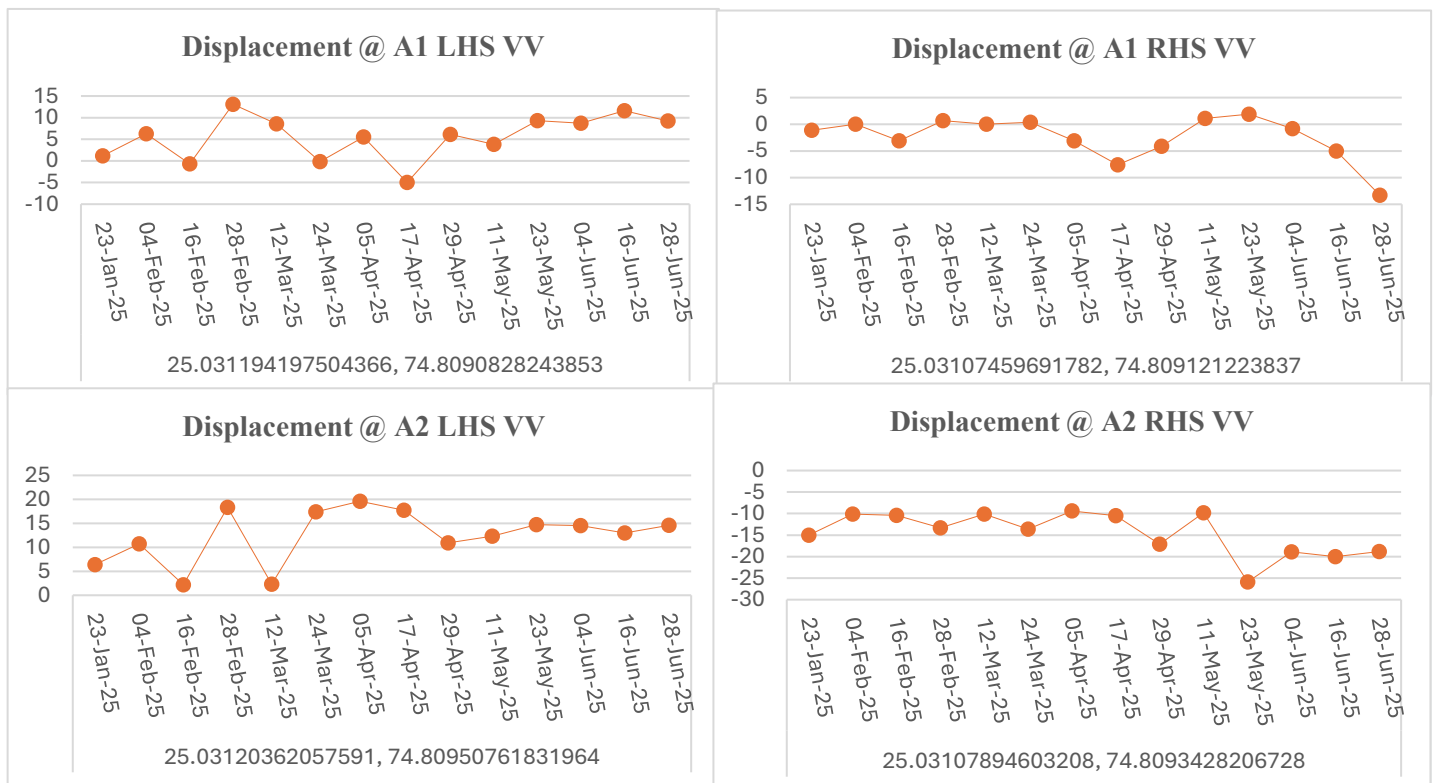
MNB – CH-929+030

Length – 45.9M

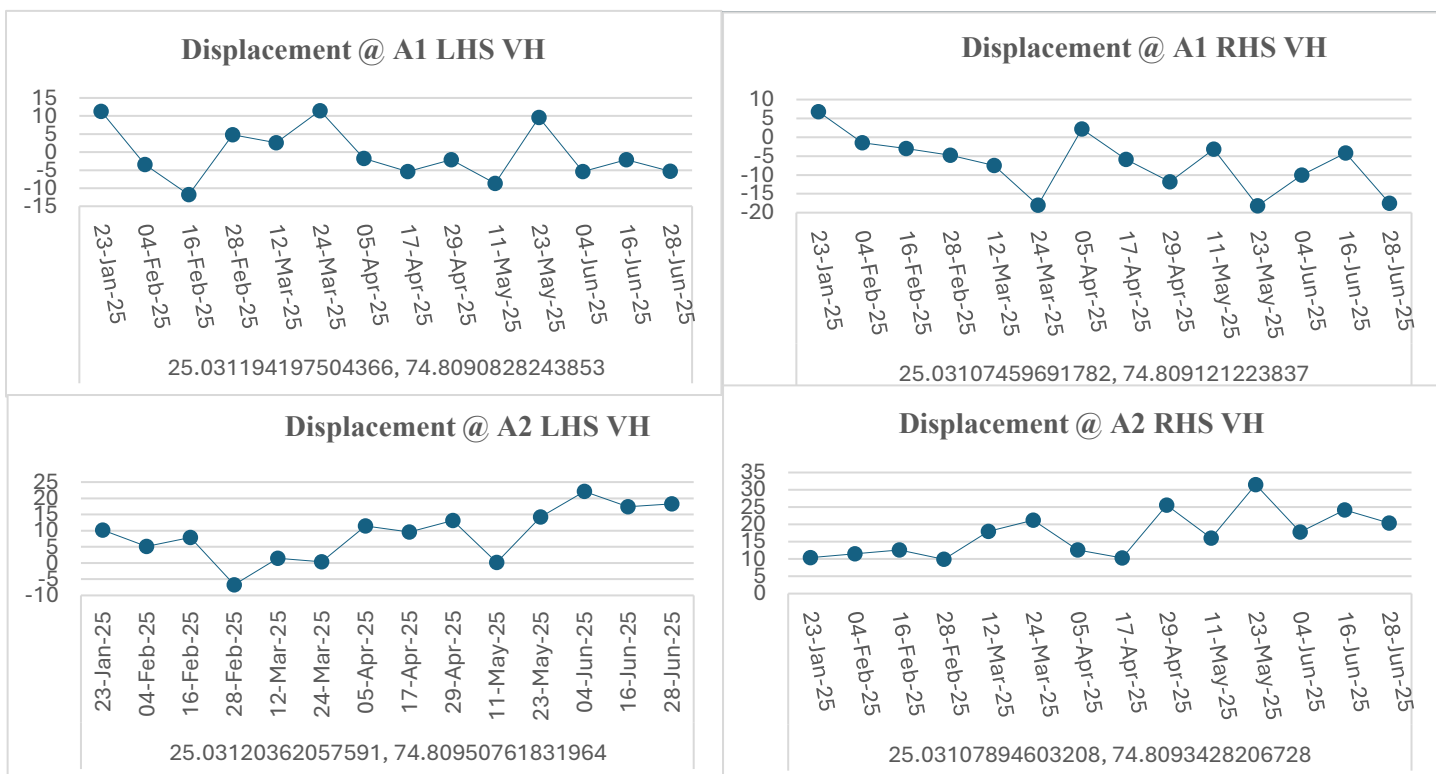


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

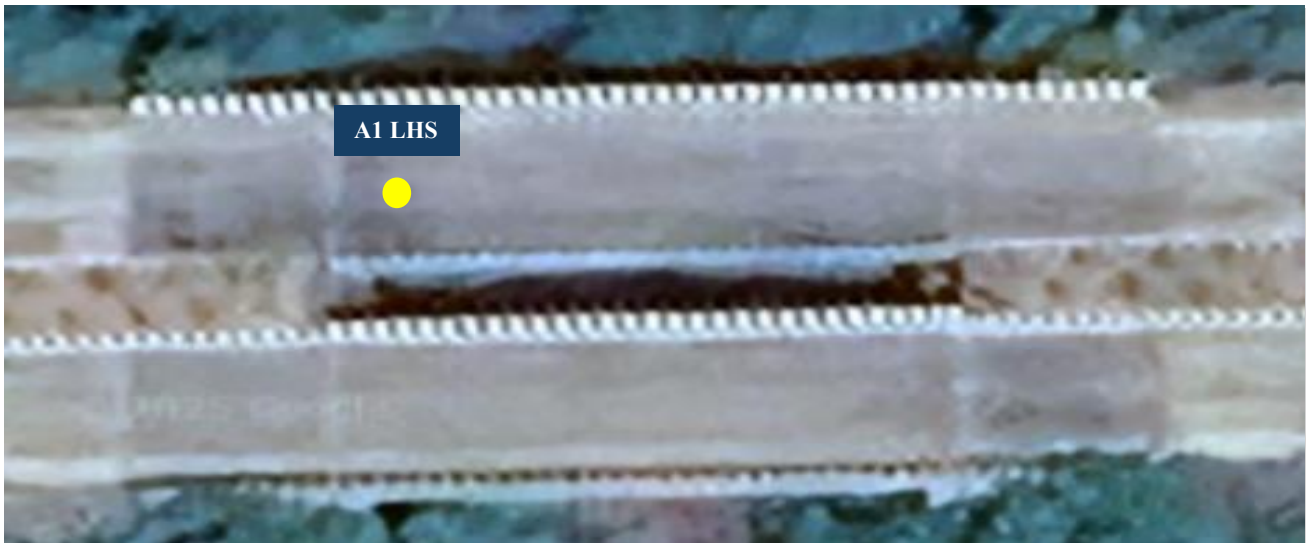
3.1 VV Displacement From 23 Jan 2025 to 28 June 2025



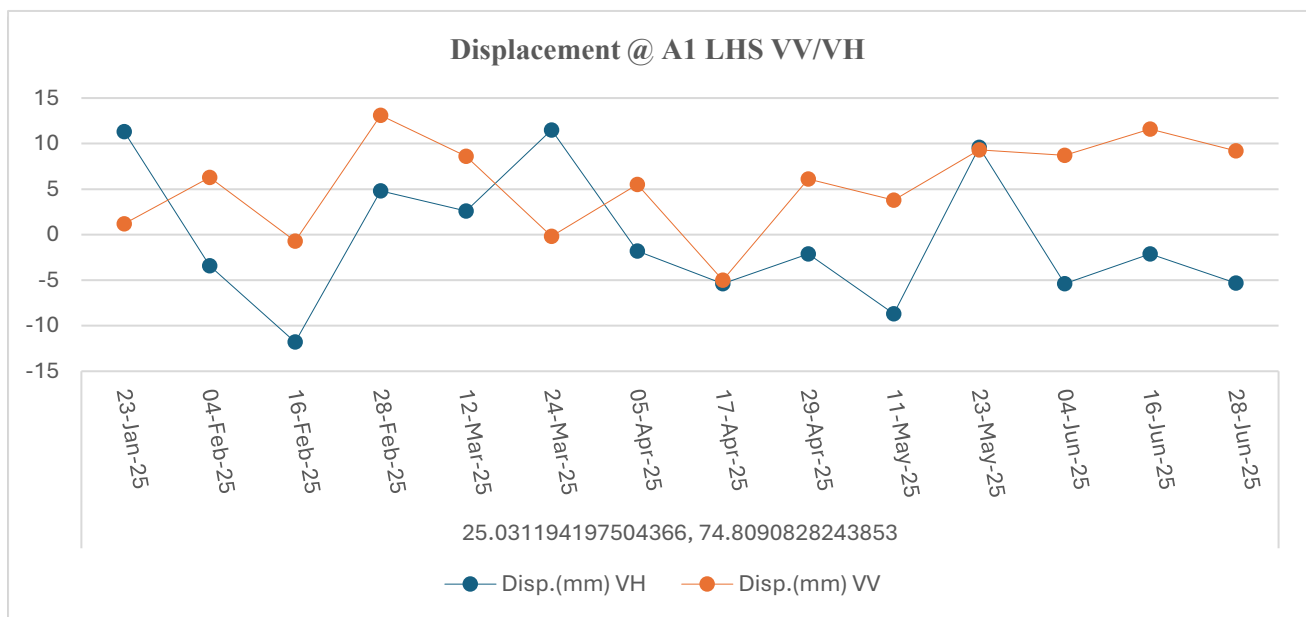
3.2 VH Displacement from 23 Jan 2025 to 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.031194197504366, 74.8090828243853)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** The vertical displacement shows moderate variability over the monitoring period, oscillating between slight uplift and downward movement. Values generally range from about –5 mm to +12 mm.
- **Horizontal displacement** Horizontal displacement exhibits comparatively higher fluctuations than vertical movement. The readings alternate between negative and positive values, approximately –12 mm to +10 mm.

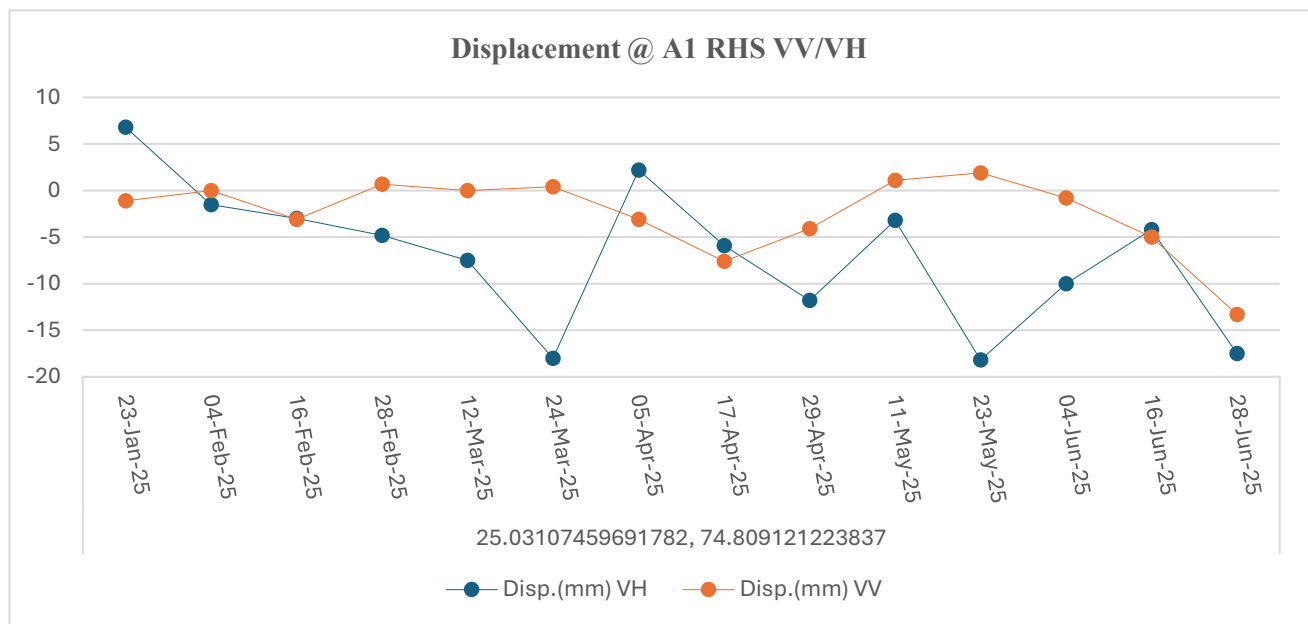
Critical Events & Anomalies:

- A noticeable negative horizontal excursion (around –12 mm) occurs during the mid-monitoring period, representing a temporary lateral shift that may be linked to transient loading conditions or thermal contraction.
- Subsequent readings demonstrate recovery toward smaller magnitudes, indicating that the movement is largely elastic and reversible rather than permanent. The vertical displacement peaks are short-lived and are not accompanied by corresponding abnormal horizontal shifts, suggesting no coupled distress mechanism.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.03107459691782, 74.809121223837)



Key Observations:

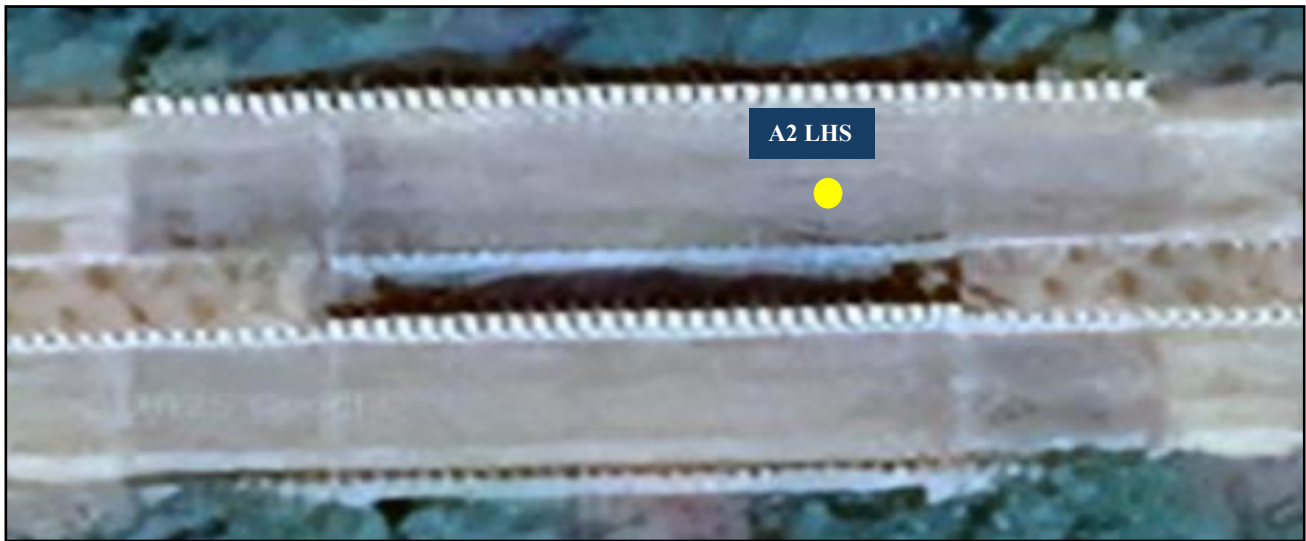
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement exhibits more pronounced variability compared to VV. Values alternate between positive and negative directions, with movements typically within ± 10 mm, but with a few notable excursions reaching approximately -18 mm.
- **Vertical displacement (VV)** The vertical displacement values show moderate fluctuations throughout the monitoring period, generally ranging between slight uplift (around $+2$ to $+3$ mm) and downward movements reaching approximately -12 to -15 mm.

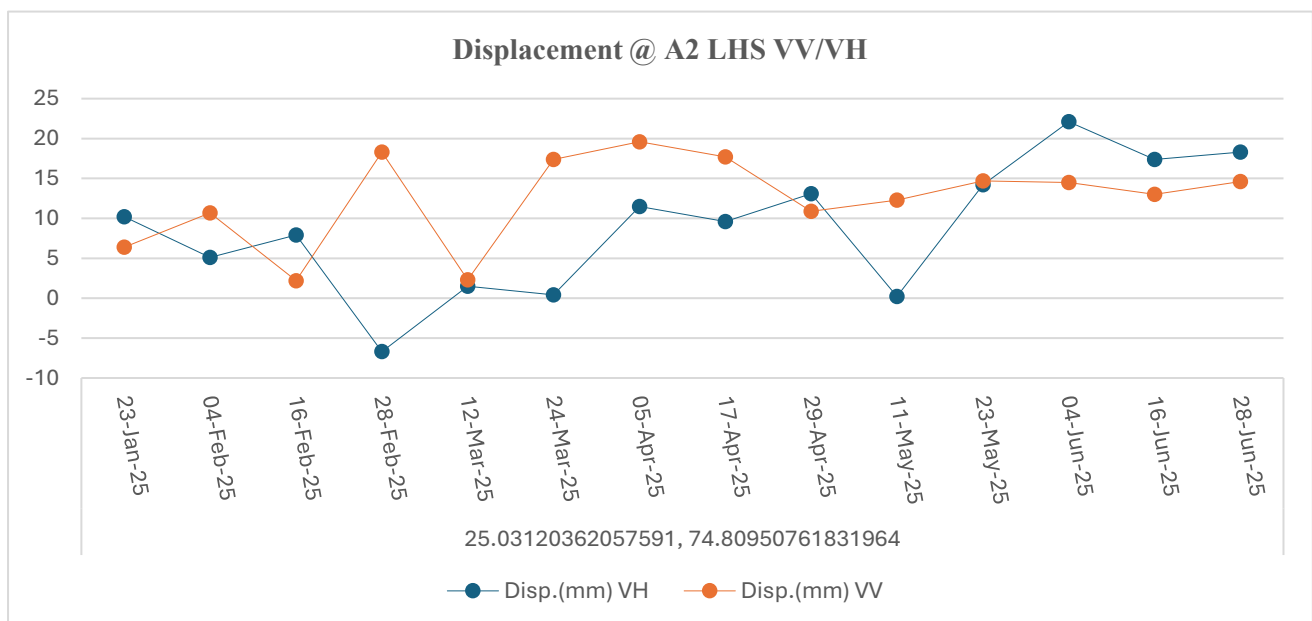
Critical Events & Anomalies:

- A distinct negative peak in VH (around -17 to -18 mm) appears during mid and late portions of the monitoring period, representing short-term lateral shifts.
- VV trends do not show abrupt or sustained drops corresponding to these VH peaks, implying that lateral movements are not coupled with significant vertical instability

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.03120362057591, 74.80950761831964)



Key Observations:

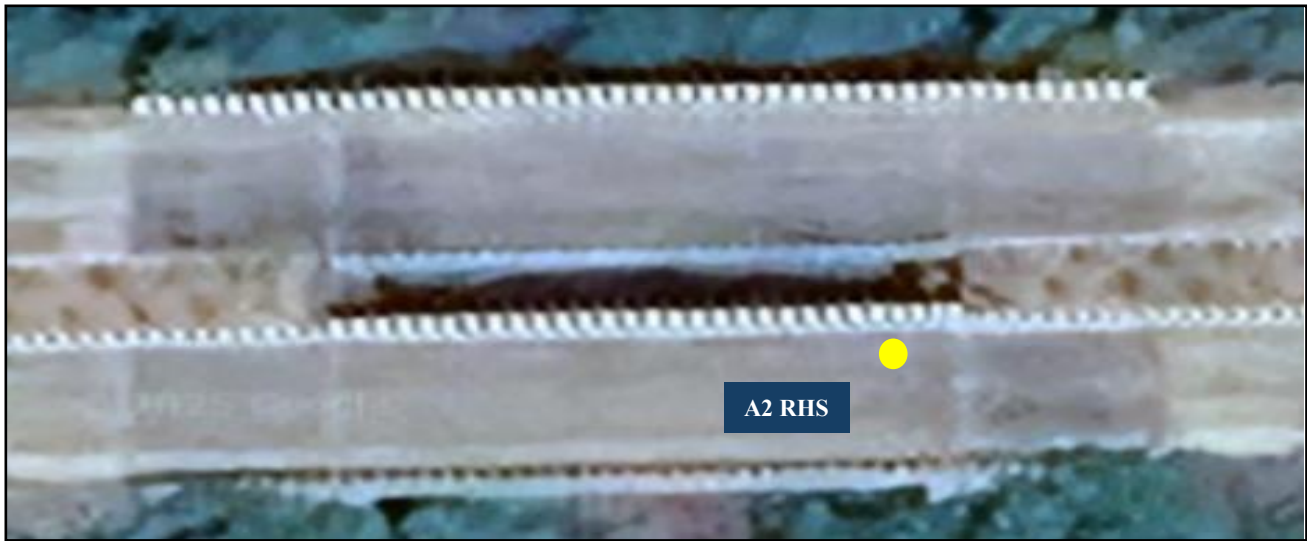
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement values show moderate fluctuations, generally remaining within a band of approximately +5 mm to +20 mm across the monitoring period.
- **Horizontal Displacement (VH)** Horizontal displacement exhibits more pronounced variability compared to vertical movement, ranging approximately from -6 mm to +22 mm

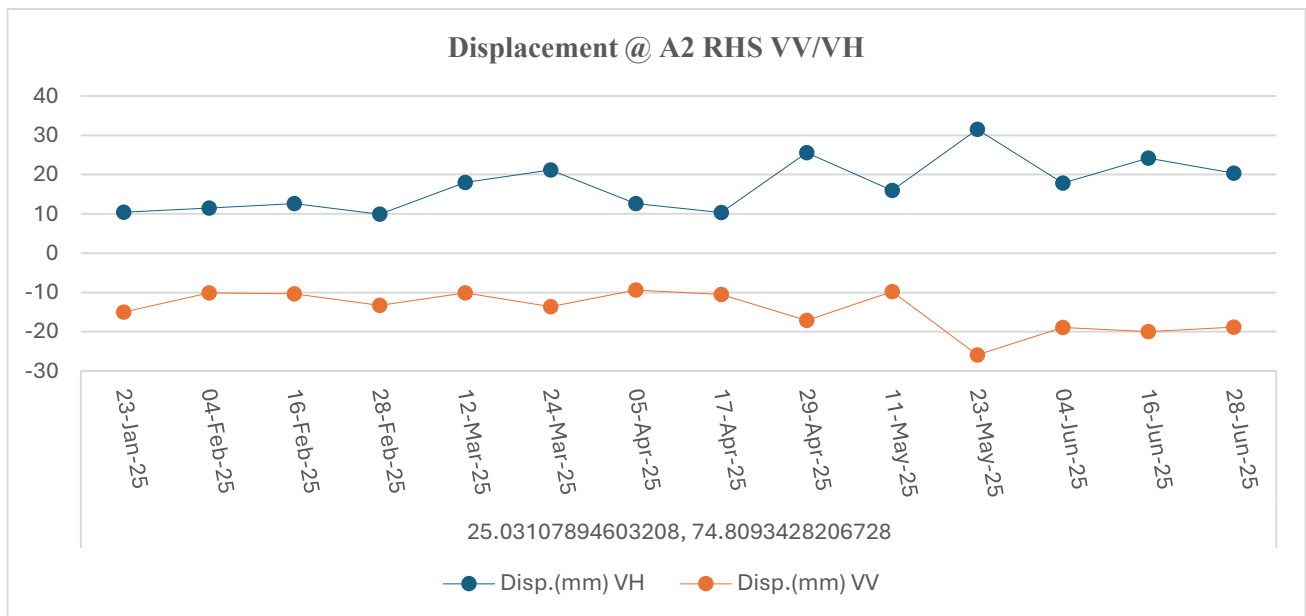
Critical Events & Anomalies:

- A short-duration negative horizontal excursion (around -6 mm) appears during the early monitoring phase, which is consistent with temporary load redistribution or environmental influence, not structural distress.
- The subsequent recovery and sustained positive horizontal displacement confirm elastic structural behaviour rather than permanent deformation.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.03107894603208, 74.8093428206728)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical displacement (VV)** Vertical displacement values show moderate fluctuations throughout the monitoring period, generally remaining within a controlled range.
- **Horizontal displacement (VH)** Horizontal displacement exhibits more pronounced variability compared to vertical movement. Early monitoring stages show alternating positive and negative shifts, reflecting reversible lateral movement rather than permanent drift.

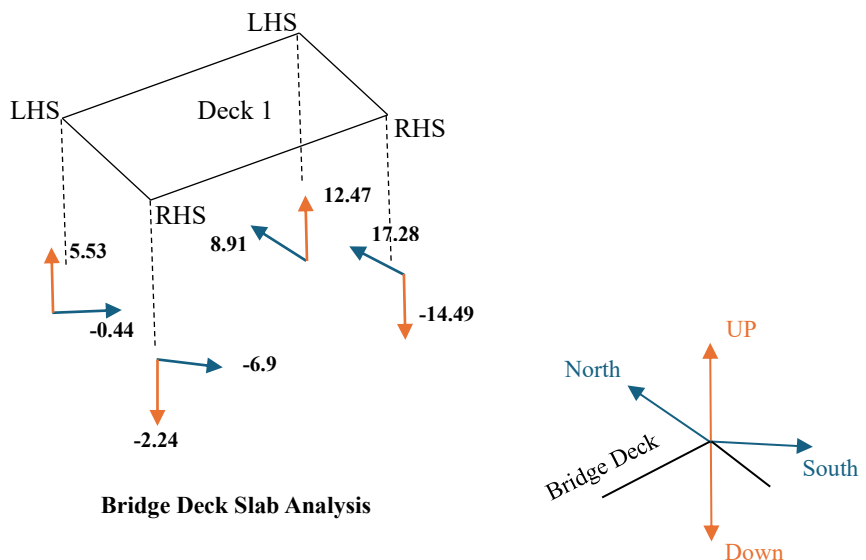
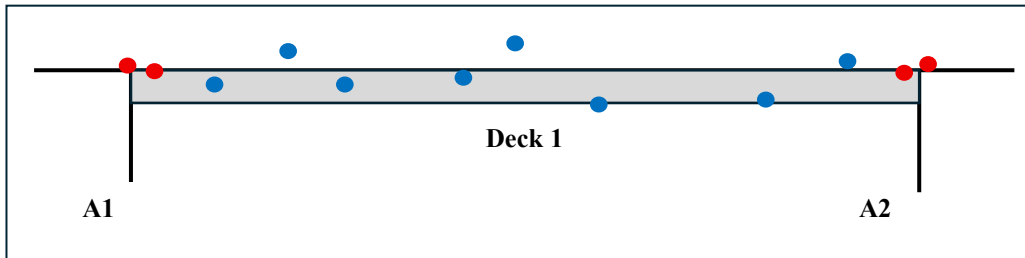
Critical Events & Anomalies:

- Isolated short-term displacement spikes are observed but are not persistent, indicating transient loading or environmental effects rather than structural distress.
- No abrupt discontinuities, step changes, or progressive divergence are detected in either vertical or horizontal components.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The deck slab exhibits noticeable differential vertical movement between LHS and RHS monitoring points, with vertical displacements ranging approximately from -14.49 mm (downward) to $+12.47$ mm (upward). The RHS of the deck shows significant downward displacement at one location, while another RHS point records pronounced upward movement, indicating localized uplift-settlement interaction.

Horizontal Displacement Behaviour:

- Horizontal displacements show clear directional variability across the deck slab. The LHS exhibits minimal horizontal movement (-0.44 mm), indicating relatively stable lateral restraint at that location.

CONCLUSION:

- The combined large vertical displacement range and opposing horizontal displacement directions indicate that Deck 1 is undergoing torsional deformation rather than rigid-body translation. The deck response suggests rotation and warping about its longitudinal axis, potentially influenced by eccentric loading, thermal effects, or differential bearing behaviour. While individual displacement magnitudes remain within moderate structural limits, the asymmetry in response warrants continued monitoring to verify whether the deformation pattern is progressive or stabilizing.